Marketing Campaign Performance

& Attribution Analytics — Executive Summary

Business Problem

Which channels actually drive conversions, what is the true cost of acquiring a customer per channel, and how should budget be reallocated to improve return on investment? This analysis combines campaign-level performance data (200,000 campaigns across 5 paid channels, full calendar year) with customer-level touchpoint data (2,847 customer journeys, 10,000 individual touchpoints across 6 channels) to answer these questions with a rigorous, auditable methodology.

Scope & Methodology

• **Data engineering:** SQL (PostgreSQL) and Python pipeline — cleaning, channel standardization, KPI derivation
• **Attribution:** four independent models (Last-Touch, First-Touch, Linear, Markov Chain removal-effect), statistically compared via chi-square test
• **Segmentation:** K-Means clustering on campaign behavioral features
• **Statistical testing:** two-proportion z-test (A/B), time-lag cross-correlation
• **Delivery:** 5-page interactive Power BI dashboard with a native DAX-driven budget reallocation simulator

A Note on the Data

The underlying datasets exhibit low variance across channels (ROI ranges 4.99–5.01; customer journey transitions are near-uniform across channels). This is stated plainly throughout this report rather than overstated: several results below are reported as "no significant difference found" rather than a manufactured story, since that is what rigorous analysis of this dataset actually supports. Where methodology would produce a sharper signal on a less uniform dataset, that is noted explicitly.

Key Performance Indicators

All figures below are computed from the full 200,000-campaign dataset (full calendar year), verified via both Python and live Power BI DAX evaluation.

Total Spend	Total Revenue	ROAS	Blended CAC	Conversion R
$2.50B	$15.02B	6.01	$284.04	8.01%

CAC & LTV by Channel

Channel	Naive CAC	Markov-Adj. CAC	LTV	CAC:LTV Ratio	Status
Direct Traffic	$281.93	$287.02	$1,694.67	0.169	Healthy
Display Ads	$284.75	$288.06	$1,708.81	0.169	Healthy
Email	$283.34	$278.25	$1,699.61	0.164	Healthy
Search Ads	$285.22	$272.26	$1,713.55	0.159	Healthy
Social Media	$284.51	$287.84	$1,708.89	0.168	Healthy

No channel shows CAC approaching LTV; all sit at a healthy ~6:1 LTV-to-CAC ratio. This is consistent with the dataset's low cross-channel variance noted above.

Top Insights

1. Attribution methodology has minimal practical impact here

Four independent attribution models (Last-Touch, First-Touch, Linear, Markov Chain removal-effect) converge to a narrow 15–18% credit band per channel. A chi-square test of homogeneity comparing Last-Touch vs. First-Touch credit distributions confirms this statistically (χ^2=2.03, p=0.845 — not significant). Channel investment decisions in this dataset should not hinge on which attribution model is used.

2. A high-risk campaign segment is not channel- or market-specific

K-Means clustering on campaign behavioral features (conversion rate, ROI, CTR, engagement, CAC) isolated a high-risk segment: **17,133 campaigns (8.6% of all campaigns), $245.7M in spend, blended CAC of $2,377 — 8.4x the $284 portfolio average.** A follow-up breakdown found no concentration by channel, location, customer segment, or campaign type (all within 0.2–0.5 percentage points of the 8.6% baseline rate). The driver is campaign-level underperformance, not a systemic channel or market issue.

3. Search vs. Email: statistically real, practically negligible

A two-proportion z-test comparing Search Ads and Email conversion rates found a statistically significant difference (p=0.0003) — but the actual gap is 0.03 percentage points (8.01% vs. 7.98%), detectable only because of the very large sample size (22M+ clicks per group). This does not justify a budget shift on its own.

4. Budget reallocation offers limited upside at current margins

Simulating a shift from the lowest-ROI channel (Display Ads, 4.99) to the highest (Direct Traffic, 5.01) projects only 0.02–0.08% revenue lift, even at a 50% budget shift — expected, given how close channel ROI is across the board.

5. Spend and conversions move together within the same month

Time-lag correlation analysis found strong same-month correlation (r=0.86–0.98) across all channels, with no reliable evidence of delayed-impact channels in this 12-month window.

Business Recommendations

1. Implement per-campaign monitoring, not blanket channel cuts

Since the $245.7M high-risk segment is not concentrated in any channel, location, or audience, set early-warning thresholds on individual campaign CTR and CAC rather than reducing budget for an entire channel.

2. Do not switch attribution models expecting a different answer

All four models statistically agree (p=0.845). Resources are better spent on the high-risk segment above than on attribution methodology debates for this dataset.

3. Treat the Search vs. Email result as informational, not actionable

The 0.03-point difference is real but too small to justify reallocating budget; continue monitoring rather than acting on this alone.

4. Budget reallocation is directionally correct but low-impact today

Shifting toward Direct Traffic is the right direction, but expect low single-digit-basis-point revenue impact, not a dramatic swing, given current channel ROI parity.

5. Prioritize root-causing the high-risk segment's campaign-level drivers

The next highest-value analysis is understanding what makes an individual campaign underperform (creative, targeting precision, timing) since the cause is not categorical.

Expected Business Impact

The estimates below are directional, based on current-margin assumptions (constant ROI as spend scales), and explicitly flagged as estimates rather than guarantees — consistent with how each figure was derived and validated during this analysis.

Action	Basis	Projected Impact
Segment-level monitoring on high-risk campaigns	$245.7M in spend flowing through the high-risk segment	Addressable spend base for efficiency improvemen
Budget shift: Display Ads to Direct Traffic (up to 50%)	Validated DAX/Python simulation, constant-ROI assumption	+0.02% to +0.08% revenue lift
Continued use of Markov-based CAC for channel decisions	More accurate than last-touch, though the gap is modest here	Improved decision accuracy, low variance in this da

Methodology & Data Notes

• Revenue derived as Spend × (1 + ROI) since the source data provided ROI but no direct revenue figure
• Campaign-level and touchpoint-level datasets joined at the channel-name level (documented mapping), not by shared customer ID
• CAC denominator (conversions) is a derived estimate (conversion rate × clicks), not a directly recorded count
• Full technical methodology, including every documented assumption and correction made during development, is available in the project's DECISIONS.md log

---